

Q.1

Solve the following linear programming problem graphically :

Maximise $Z = 20x + 30y$

Subject to the constraints :

$$x + y \leq 80$$

$$2x + 3y \geq 100$$

$$x \geq 14$$

$$y \geq 14$$

Q.2

Solve the following linear programming problem graphically :

Maximize $Z = 8x + 9y$

Subject to the constraints :

$$2x + 3y \leq 6$$

$$3x - 2y \leq 6$$

$$y \leq 1$$

$$x \geq 0, y \geq 0$$

Q.3

Solve the following linear programming problem graphically :

Minimise $Z = x - 5y$

subject to the constraints :

$$x - y \geq 0$$

$$-x + 2y \geq 2$$

$$x \geq 3, y \leq 4, y \geq 0$$

Q.4

Solve the following Linear Programming Problem graphically :

Minimise $Z = 3x + 5y$

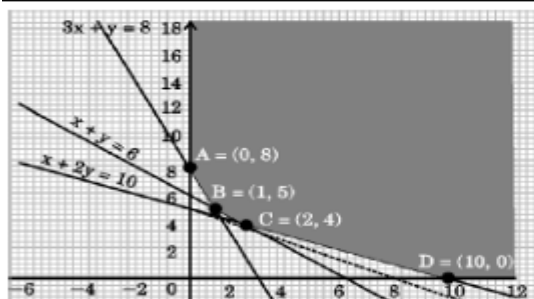
subject to the constraints

$$x + 2y \geq 10$$

$$x + y \geq 6$$

$$3x + y \geq 8$$

$$x, y \geq 0$$



Corner Point	Value of $Z = 3x + 5y$
A(0, 8)	40
B(1, 5)	28
C(2, 4)	26
D(10, 0)	30

$3x + 5y < 26$ has no common region with the feasible region.

$\therefore Z_{\min} = 26$

Q.5

The corner points of the feasible region of a Linear Programming Problem are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). If $Z = ax + by$; ($a, b > 0$) be the objective function, and maximum value of Z is obtained at (0, 2) and (3, 0), then the relation between a and b is :

(A) $a = b$

(B) $a = 3b$

(C) $b = 6a$

(D) $3a = 2b$

Q.6

Assertion (A) : In a Linear Programming Problem, if the feasible region is empty, then the Linear Programming Problem has no solution.

Reason (R) : A feasible region is defined as the region that satisfies all the constraints.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

Q.7

Solve the following linear programming problem graphically :

Maximise $Z = x + 2y$

Subject to the constraints :

$$x - y \geq 0$$

$$x - 2y \geq -2$$

$$x \geq 0, y \geq 0$$

Q.8

The corner points of the feasible region determined by some system of linear inequations, are (0, 0), (5, 0), (3, 4) and (0, 5). Let $Z = ax + by$, where $a, b > 0$. Find the condition on a and b so that the maximum of Z occurs at both points (3, 4) and (0, 5).

Q.9

The corner points of the feasible region in graphical representation of a L.P.P. are (2, 72), (15, 20) and (40, 15). If $Z = 18x + 9y$ be the objective function, then

- (A) Z is maximum at (2, 72), minimum at (15, 20)
- (B) Z is maximum at (15, 20) minimum at (40, 15)
- (C) Z is maximum at (40, 15), minimum at (15, 20)
- (D) Z is maximum at (40, 15), minimum at (2, 72)

Q.10

If the feasible region of a linear programming problem with objective function $Z = ax + by$, is bounded, then which of the following is correct ?

- (A) It will only have a maximum value.
- (B) It will only have a minimum value.
- (C) It will have both maximum and minimum values.
- (D) It will have neither maximum nor minimum value.

Q.11

A factory produces two products X and Y. The profit earned by selling X and Y is represented by the objective function $Z = 5x + 7y$, where x and y are the number of units of X and Y respectively sold. Which of the following statement is correct ?

- (A) The objective function maximizes the difference of the profit earned from products X and Y.
- (B) The objective function measures the total production of products X and Y.
- (C) The objective function maximizes the combined profit earned from selling X and Y.
- (D) The objective function ensures the company produces more of product X than product Y.

Q.12

Assertion (A) : Every point of the feasible region of a Linear Programming Problem is an optimal solution.

Reason (R) : The optimal solution for a Linear Programming Problem exists only at one or more corner point(s) of the feasible region.

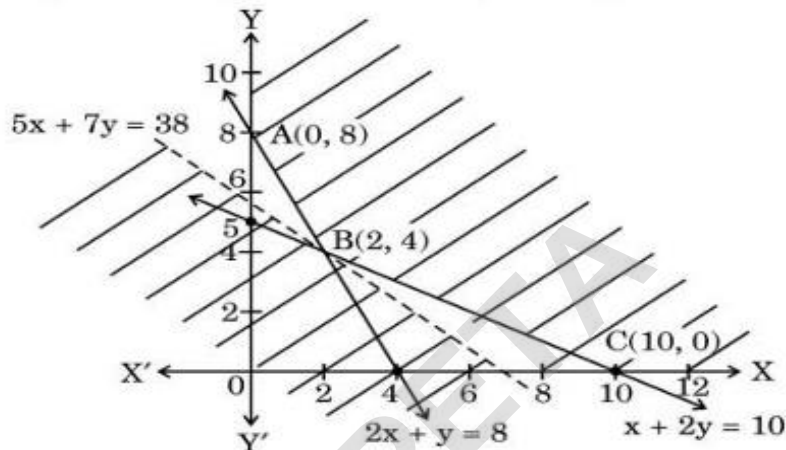
Q.13

Of all the points of the feasible region, for maximum or minimum values of the objective function, the point lies :

- (A) inside the feasible region
- (B) at the boundary line of the feasible region
- (C) at the corner points of the feasible region
- (D) at the coordinate axes

Q.14

Assertion (A) : The shaded portion of the graph represents the feasible region for the given Linear Programming Problem (LPP).



$$\text{Min } Z = 50x + 70y$$

subject to constraints

$$2x + y \geq 8, \quad x + 2y \geq 10, \quad x, y \geq 0$$

$Z = 50x + 70y$ has a minimum value = 380 at $B(2, 4)$.

Reason (R) : The region representing $50x + 70y < 380$ does not have any point common with the feasible region.

Q.15

In the Linear Programming Problem (LPP), find the point/points giving maximum value for $Z = 5x + 10y$

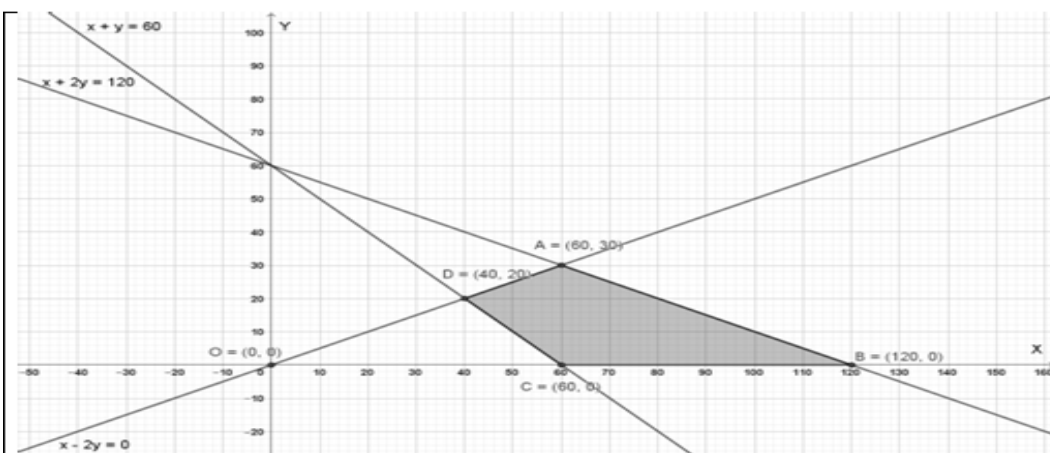
subject to constraints

$$x + 2y \leq 120$$

$$x + y \geq 60$$

$$x - 2y \geq 0$$

$$x, y \geq 0$$



Corner Points	Value of Z
A (60, 30)	600
B (120, 0)	600
C (60, 0)	300
D (40, 20)	400

Since Z is maximum on points A and B

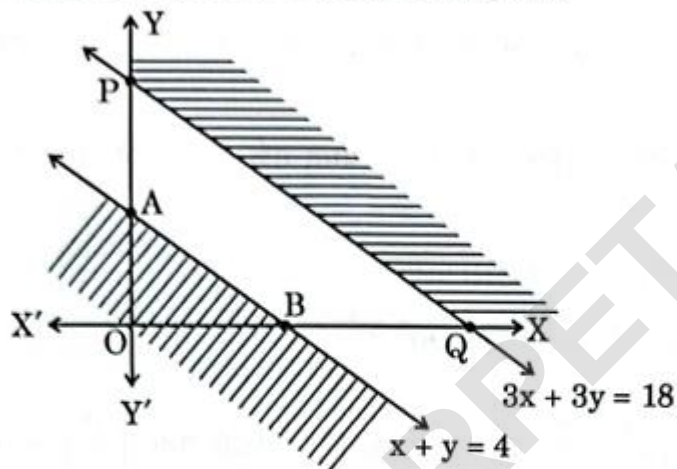
Hence all points lying on segment AB give maximum Z.

Q.16

In a Linear Programming Problem (LPP), the objective function $Z = 2x + 5y$ is to be maximised under the following constraints :

$$x + y \leq 4, \quad 3x + 3y \geq 18, \quad x, y \geq 0$$

Study the graph and select the correct option.



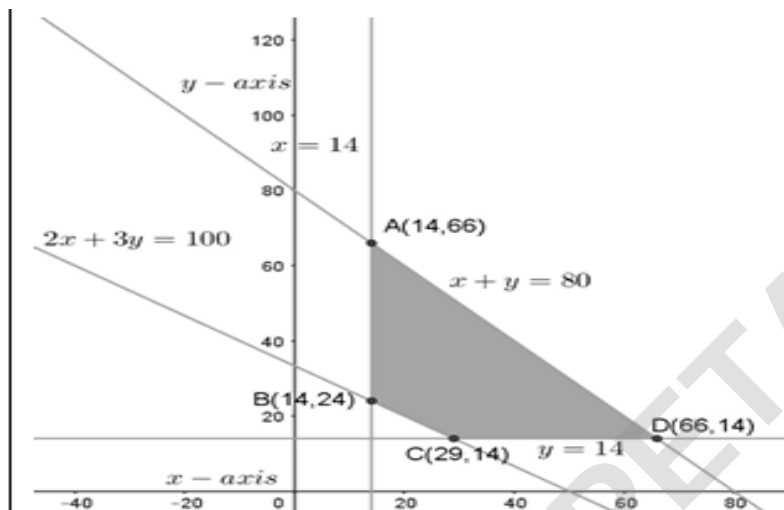
(Note : The figure is not to scale)

The solution of the given LPP :

- (A) lies in the shaded unbounded region.
- (B) lies in ΔAOB .
- (C) does not exist.
- (D) lies in the combined region of ΔAOB and unbounded shaded region.

HINTS.

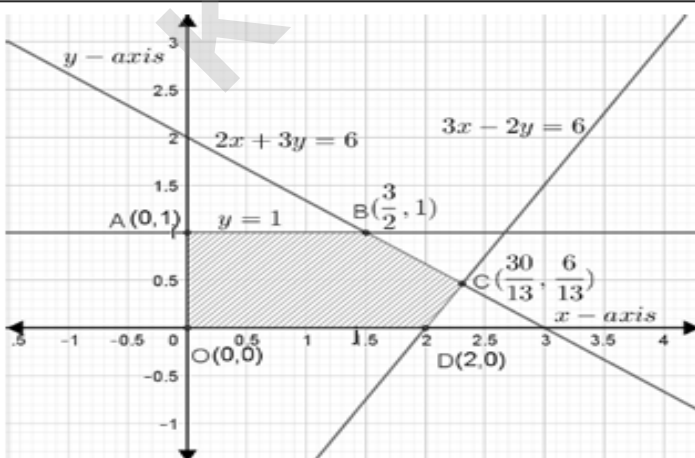
Q.1



Corner Point	Value of $Z = 20x + 30y$
A(14,66)	2260
B(14,24)	1000
C(29,14)	1740
D(66,14)	1000

Max(Z) = 2260

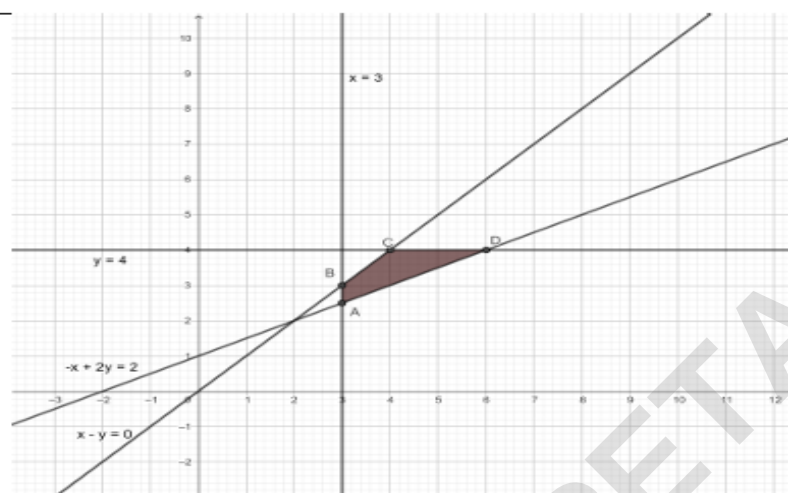
Q.2



Corner Point	Value of $Z = 8x + 9y$
A(0,1)	9
B($\frac{3}{2}$, 1)	21
C($\frac{30}{13}$, $\frac{6}{13}$)	$\frac{294}{13}$
D(2,0)	16
O(0,0)	0

Max(Z) = $\frac{294}{13}$

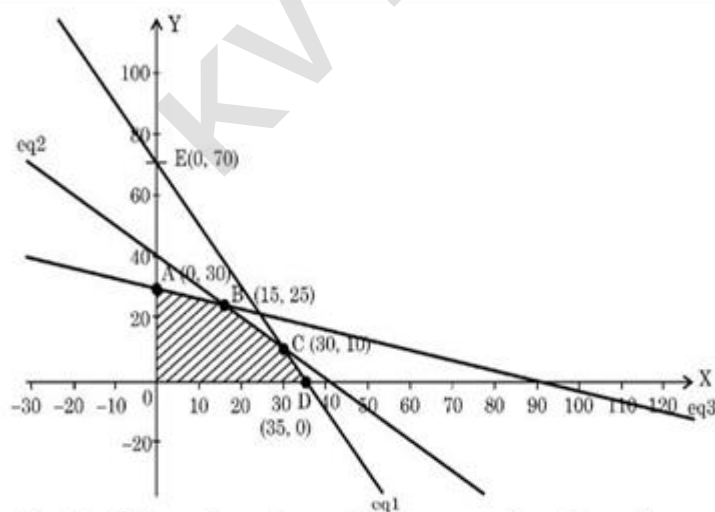
Q.3



Corner point	Value of $Z = x - 5y$
A (3, 2.5)	-9.5
B (3, 3)	-12
C (4, 4)	-16
D (6, 4)	-14

The minimum value of Z is -16, which is attained at $x = 4$, $y = 4$.

Q.



The feasible region along with corner points for a linear programming problem are shown in the graph. Write all the constraints for the given linear programming problem.

The equation of the line AB is

$$y - 30 = \frac{25-30}{15-0} (x - 0)$$

$$\Rightarrow x + 3y = 90$$

The equation of the line BC is

$$y - 10 = \frac{10-25}{30-15} (x - 30)$$

$$\Rightarrow x + y = 40$$

The equation of the line CD is

$$y - 10 = \frac{0-10}{35-30} (x - 30)$$

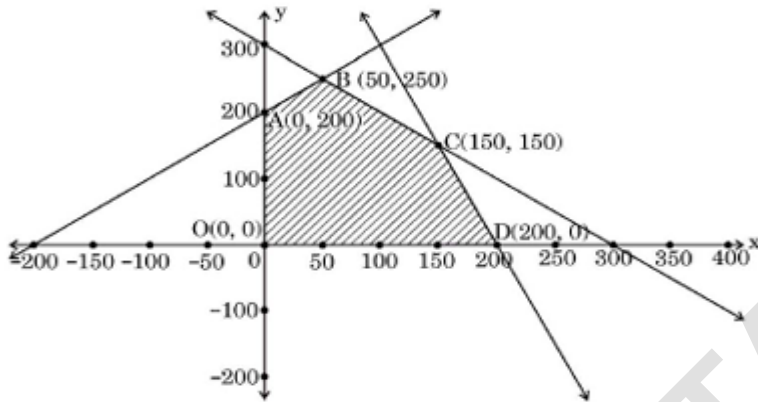
$$\Rightarrow 2x + y = 70$$

Hence, the constraints are

$$x + 3y \leq 90, x + y \leq 40, 2x + y \leq 70$$

$$x \geq 0, y \geq 0$$

Q.5



For the given graph of a Linear Programming Problem, write all the constraints satisfying the given feasible region.

Equation of AB:

$$y - 200 = \frac{250 - 200}{50 - 0}(x - 0) \text{ i.e. } -x + y = 200$$

Equation of BC:

$$y - 250 = \frac{150 - 250}{150 - 50}(x - 50) \text{ i.e. } x + y = 300$$

Equation of CD:

$$y - 0 = \frac{0 - 150}{200 - 150}(x - 200) \text{ i.e. } 3x + y = 600$$

Required constraints are:

$$-x + y \leq 200$$

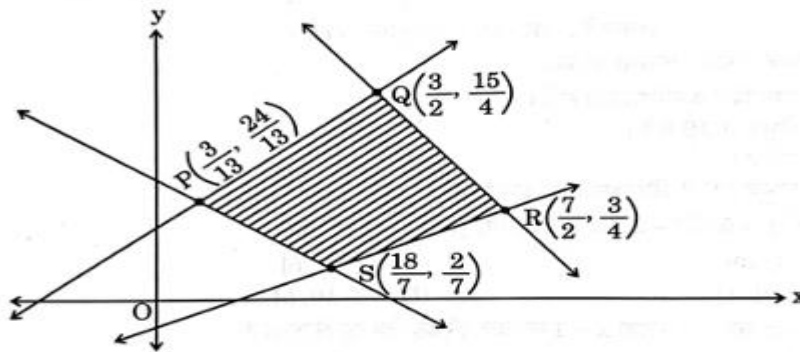
$$x + y \leq 300$$

$$3x + y \leq 600$$

$$x \geq 0, y \geq 0$$

Q.6

For a Linear Programming Problem (LPP), the given objective function is $Z = x + 2y$. The feasible region PQRS determined by the set of constraints is shown as a shaded region in the graph.



(Note : The figure is not to scale)

$$P = \left(\frac{3}{13}, \frac{24}{13}\right), Q = \left(\frac{3}{2}, \frac{15}{4}\right), R = \left(\frac{7}{2}, \frac{3}{4}\right), S = \left(\frac{18}{7}, \frac{2}{7}\right)$$

Which of the following statements is correct ?

(A) Z is minimum at $S\left(\frac{18}{7}, \frac{2}{7}\right)$

(B) Z is maximum at $R\left(\frac{7}{2}, \frac{3}{4}\right)$

(C) (Value of Z at P) > (Value of Z at Q)

(D) (Value of Z at Q) < (Value of Z at R)

Q.7

Consider the Linear Programming Problem, where the objective function $Z = (x + 4y)$ needs to be minimized subject to constraints

$$2x + y \geq 1000$$

$$x + 2y \geq 800$$

$$x, y \geq 0.$$

Draw a neat graph of the feasible region and find the minimum value of Z .



Correct Graph and shading:

Corner points

$(800, 0)$

$(400, 200)$

$(0, 1000)$

Value of Z

800

1200

4000

$x + 4y < 800$ has no region common with feasible region, hence 800 is minimum